Supply Chain Management: A Multi-Agent System for Collaborative Production Planning

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Abstract

The need for coordinated material flow in supply chains is essential for the competitiveness of manufacturing firms. A hierarchical coordination is not applicable for each supply chain. A collaboration between entities is required to meet future challenges. Besides appropriate planning concepts an IT solution is a major challenge. Multi-Agent Systems (MAS) are an enabler for a decentralized coordination process. The evolvement of appropriate planning methods makes MAS interesting for collaborative Supply Chain Management (SCM). Nevertheless, the user should be seen as the supply chain engineer, holding the knowledge to create the collaborative supply chain processes. Therefore, the user has to be integrated in the IT solution. A configurable MAS, based on generic components, satisfies this requirement so that MAS can easily be adapted to the changing processes and environment by the user. This paper presents a concept of a configurable MAS and a prototype of a MAS-Editor for collaborative production planning.

1. Introduction

Production networks evolve due to the outsourcing strategies and core business concentration of today's companies. This production network or supply chain needs to be coordinated to fulfill the customer's demands at the end of the chain. The supply chain is exposed to a dynamic changing environment, as a result of the unsteady demand for products and organizational re-structuring. Supply Chain Management (SCM), on the operational level, requires a coordination that keeps the autonomy of the entities in their local information, planning methods and decision making. Multi-Agent Systems (MAS) are an enabler for a decentralized coordination process. Existing planning methods have to be improved as well as new planning concepts worked out, which fit to decentralized coordination processes. Furthermore the user should be seen as a privileged member, holding the knowledge to create and shape a collaborative supply chain management.

This paper is structured as follows: The second section introduces collaborative SCM and points out the motivation for agent technology as an appropriate IT-System. In the third section a MAS concept for collaborative production planning is raised. A collaboration process is exemplarily shown in section four. A prototype of a MAS editor is presented in section five.

2. Collaborative Production Planning as part of Supply Chain Management

2.1. Overview

Supply Chain Management (SCM) aims to manage production networks or supply chains by an integrated, process orientated planning and control of the material and information flow along the whole supply chain [3,9]. Existing SCM-Systems are based on a hierarchical coordination within the supply chain, where one focal entity shapes and controls the whole supply chain.

Collaborative Supply Chain Management (CSCM) tries to overcome this hierarchical coordination. The relationship between the entities is bilateral; both parties have the power to shape its nature and future direction over time. Mutual commitment to the future and a balanced power relationship are essential [13]. Information exchange enables cooperated coordination to optimize the supply chain [2]. The

information exchange should enable the synchronization of the material flow throughout the whole supply chain.

2.2. Motivation

Given a serial production, where skeleton contracts define restrictions of exchange of goods in regard to quantity, quality, prices etc., a different approach for supply chain wide production planning seems appropriate. There will be no separation of the inter-organizational collaboration and the intra-organizational production planning. Collaborative production planning has to be realized without this differentiation. The supply chain will be fragmented into individual, autonomous entities, able to fulfill the required coordination and local planning by themselves. The user has to be considered as the supply chain engineer, who has the knowledge to form the relationship and define collaboration processes with appropriate planning processes. Without the acceptance and integration of the user the IT solution will not succeed.

The enabler of an IT solution can be found in the field of Distributed Artificial Intelligence (DAI). The research domain of Multi-Agent-Systems (MAS) is part of it. Agents are characterized as reactive, proactive and autonomous [20]. Furthermore, they are able to interact and communicate, consisting of a social behavior. Requirements of collaborative production planning are met by agents. Nevertheless, the MAS has to be designed in a way that integrates the user as the supply chain engineer. The behavior of the individual agents cannot be defined at design time. Considering a generic approach, the MAS has to be initialized by a limited number of agent types, which have to be individualized by the user.

3. Agent-based collaborative production planning

In this section relevant issues of the agent solution are discussed. The agent model is shortly introduced in section 3.1, followed by the decomposition of the agents regarding the supply chain in section 3.2. The coordination concept of the agents is discussed in section 3.3. The local planning is subject of section 3.4 before the configuration concept is presented in section 3.5.

3.1. Agent model

The BDI-model (Belief-Desire-Intention) is an often used theoretical concept of an agent model [19]. This concept is also suited to be used for collaborative production planning in a supply chain. Furthermore, the BDI concept is suited to integrate the user by the ability to configure each agent.

The transformation of the theoretical model to a practical approach can be realized as follows. An agents' belief corresponds to the information he has about the world. Therefore he has to recognize his environment. Corresponding the collaborative planning, the relevant information consists of the conversation state, the plan state and the state of the ongoing or finished planning processes. The desires of an agent can be realized by tasks. A task corresponds to a goal the agent wants to achieve. The number of possible tasks depends on the functional ability of the agent. Plans correspond to the intentions of an agent. They are used to achieve the dedicated task and change the environment, i.e. the production plan and ongoing conversation. Plans consist of elemental functions which can be parameterized.

3.2. Decomposition of agents

A Multi-Agent System can be defined as a set of nodes, designated by agents that represent the objects of the system [6]. The decomposition describes the identification of the objects to be represented by an agent.

According to this object orientated decomposition MFERT [5] is an appropriate base for the definition of the agents. This model offers a representation of a production process, regarding the required planning tasks. It consists of 3 types of production entities: materials, processes and resources. Therefore, the agents are decomposed in material-, process- and resource-agents which will be linked according to the flow of the material and capacity respectively. Inter-organizational interfaces are not explicitly considered due to the integrated supply chain concept.

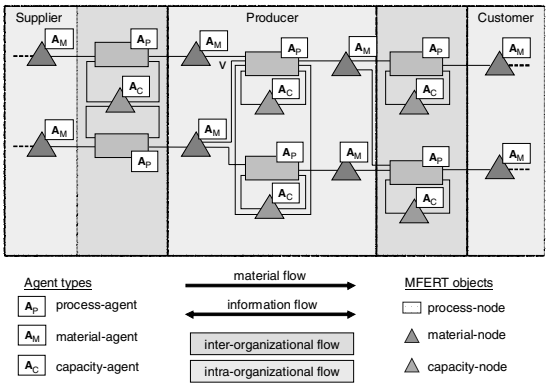


Figure 3.1 MAS representing entities of the supply chain

3.3. Coordination

The coordination is realized by an agent communication, based on speech acts that express the intent of an agent. The structure of the message follows the KQML (Knowledge Query and Manipulation Language) approach [7], whereas the content is coded in KIF (Knowledge Interchange Format) [10]. The first supporter of an agent architecture combining KIF and KQML was presented by the Knowledge Sharing Effort in 1992 [16].

The negotiation protocol itself is relatively standard and based on speech act performatives [1,4,17]. It covers the process of synchronizing the production plan. The performatives are "propose", "counter-propose", "accept", "deny" and "reject". The protocol has to be specified by the supply chain engineers and can be adapted to special requirements so that the possible sequence of the speech-act can be restricted and an efficient collaboration realized.

An example of the protocol, using a UML state chart diagram, for a material agent receiving a proposal from a process agent is showed in figure 3.2.

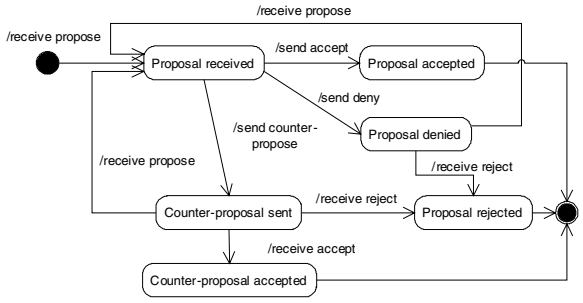


Figure 3.2 Protocol of a material agent receiving a proposal

3.4 Local Planning

The production planning can be characterized by the planning tasks and planning processes to fulfil these tasks. A systemization of the tasks and processes is necessary to enable a comprehensive description of the production planning. Starting from relevant planning events, leading to relevant states, the tasks and appropriate processes are identified for each type of production entity. The characteristics of material and resource nodes are similar so that the events, states and tasks are transferable.

The relevant events are represented by the information exchange of material supply as well as requirements and capacity respectively. Additional information e.g. priorities for stating an emergent order, etc. can be considered for the planning process.

The events are interpreted and become part of the production plan. The altered plan is checked for inconsistencies, which results in a state description. The inconsistencies arise from the defined restrictions of production entities.

The process of tasks by plans lead to events. The effect of tasks can therefore be differentiated according to their direction, i.e. upstream, downstream or limited to the node, as well as their quantity, i.e. increase or decrease of material and capacity respectively. Regarding the network structure and characteristics of the exchange of goods, plans are specified in more detail.

Focussing on serial production, the exchange of goods can be characterized by lot size and ordering interval [15]. Fixed Order Quantity relies on fixed lot sizes and variable ordering intervals. Fixed Period Requirements focus on variable lot sizes and fixed ordering intervals. Variable lot sizes and variable ordering intervals are characteristics of Discrete Ordering as well as cost orientated ordering techniques like Least Unit Cost Least Total Cost, Part Period Balancing or Wagner-Within Algorithm.

The patterns of the material flow increasing concerning single sourcing and differentiated due to the types of the characteristics of the planned material flow are shown in figure 3.3. A multiple sourcing possibility will extend the methods by a variation of taking priorities of the alternative processes into account or not.

Characteristics		Plan
lot grouping	lot size	
no	variable	period-wise increasing
yes	variable	lot increasing (forward)
yes	variable	lot increasing (backward)
yes	fixed	interval decreasing

Figure 3.3 Single sourcing patterns for material flow increasing

3.5 Configuration

As posted, the protocol is mandatory for the negotiation. The success of the negotiation depends on the reasoning, i.e. choosing the planning method to fulfil the subject of the proposal considering the current situation. This can be modelled as a knowledge base consisting of two components: a declarative one and a procedural one.

The declarative knowledge comprises the negotiation state, the current state of the production plan and the state of the ongoing planning processes. This knowledge refers to the beliefs of the agent. The procedural knowledge base contains statements how to react, given the declarative knowledge. According

to the BDI concept, the desires and intentions are related to the beliefs.

In its simplest form, an agent reacts in a direct, predetermined response to a particular event. Typically, the behaviour of reactive agents is expressed in the form: WHEN event, IF condition(s), THEN action. The scheme also fits in the BDI concept so that a rule-based agent architecture is chosen for this application.

4. Scenario of collaborative planning

The following scenario points out the interrelation of the communication and the local planning in a supply chain. The model consists of a material entrance, represented by $Agent_{IN}$, a process, represented by $Agent_{PROCESS}$, an associated resource, represented by $Agent_{RESOURCE}$ and a material exit, represented by $Agent_{OUT}$ (figure 4.1). The agents communicate according to the production structure to synchronize the decentralized production plans. The initial production plan is consistent over all entities. The intra-organizational material flow can be planned with variable lot sizes and a variable ordering intervals. The inter-organizational material flow on the upstream side has to consider a fixed ordering interval and variable lot sizes, whereas the downstream material flow depends on fixed lot sizes and variable ordering intervals.

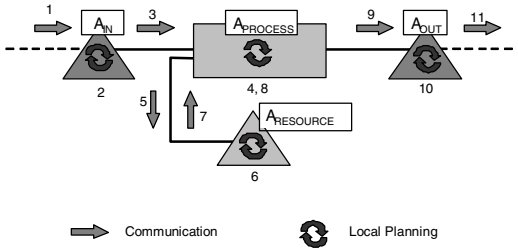


Figure 4.1 Model of the scenario

The announcement (1) of a delayed material supply by the supplier to the material entry agent (A_{IN}) initiates the collaboration process. In period 1 the incoming lot is reduced by 100, whereas each lot in period 5 and 9 is increased by 50. The overall inflow quantity does not change within the planning horizon. The reduction is fixed and cannot be negotiated. The result of the local planning (2) comes up with a fall below the minimum restriction, defined by a fixed safety stock of 50. A reduction of the safety stock would partly compensate the delay but is not taken into account. Therefore, the net supply cannot be kept as planned and a task for net supply reduction by a "lot decreasing–backward" is chosen. This reduces the lot in period 4 by 100. Following the net supply reduction, a net supply increasing wit a "lot increasing–forward" is executed to fulfill the gross requirements quantity. This leads to increased lots in period 6 and 10 by 50. The overall net supply quantity is hold. The plan of the old and new material stock is shown in figure 4.2.

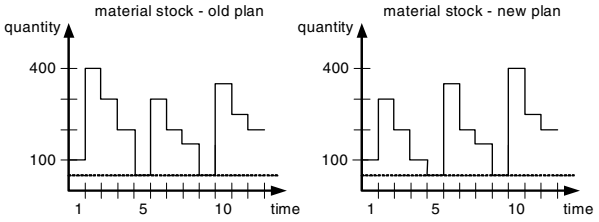


Figure 4.2 Material entry stock

The change of the net supply is sent forward to the process agent ($A_{PROCESS}$) (3). The process agent calculates new capacity requirements and communicates them to the resource agent ($A_{RESOURCE}$) (5). According to his work load (6), the resource agent is able to respond with the required capacity supply (7). The lead time counts one timeslot for each batch. If the material and capacity supplies match, the process agent communicates the changed material flow to the following material agent (A_{OUT}) (9). Depending on the stock on hand, the delay may be compensated. In case of a material shortage, a safety stock reduction may be undertaken but is not considered in this scenario. If the outflow of material can not be met as planned, the plan of the net supply has to be adjusted according to the gross supply and minimum restrictions. Due to fixed lot sizes of 300, the task net supply reduction has to be fulfilled by "ordering interval increasing". The former net supply was planned for period 2, 6, and 10. Regarding the gross supply changes, the new lots are planned for period 2, 8, and 12, as shown in figure 4.3.

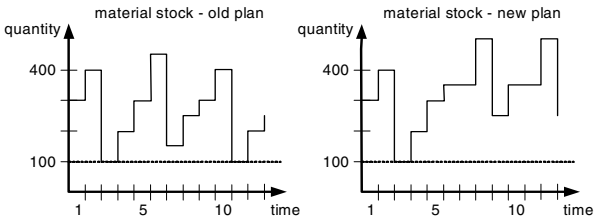


Figure 4.3 Material exit stock

The scenario showed how a plan change may spread in the supply chain. The collaborative production planning of the agents enable the

information transparency and are equipped with the ability to synchronize decentralized production plans.

5. MASCOPP Prototype

The prototype MASCOPP (Multi-Agent System for collaborative production planning) consists of an editor, to model the agent network and a Multi-Agent System runtime environment in which the agents are initialized.

The editor enables a graphical modeling of the multi-agent network according to the represented supply chain. Due to the complexity and size of the supply chain, the MAS can be modeled in fragments. The runtime environment will take care about the linking of the fragments. The editor is shown in figure 5.1. It consists of a menu bar with common features (1), such as loading and saving projects, a toolbox for choosing the type of agent to model (2), the drawing area (3) and a log viewer (4).

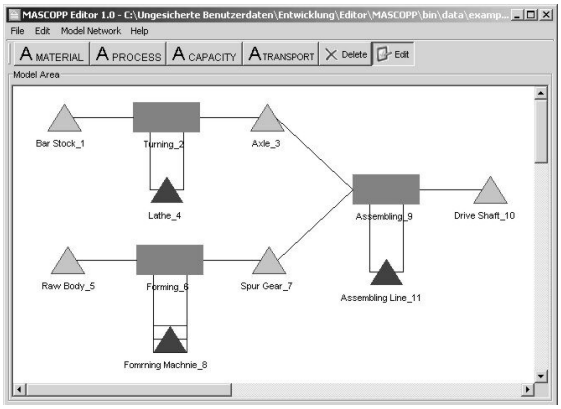


Figure 5.1 MASCOPP Editor

The properties of each agent can be specified via a dialog form. Besides the unique name, various properties like the material flow restrictions of the representing entity can be specified.

The MAS model including the properties of each agent is saved in an MAS definition XML file. This file will initiate the MAS in the runtime environment. Has the MAS been modeled in fragments, each fragment will be initiated in a separate runtime environment. The linking between the MAS fragments is realized by the runtime environment.

The behavior of each agent is specified by rules. These are formulated in the rule editor, which is also part of the MASCOPP editor. The rule base of each agent is saved in a separate XML file. The rules are evaluated by the agents at runtime so that an online adaptation is possible.

6. Summary

This paper has described collaborative production planning as part of the Collaborative Supply Chain Management. The goal was a synchronization of decentralized production plans of autonomous entities.

The enabling technology was found in the Multi-Agent Systems (MAS) as part of the field of Distributed Artificial Intelligent (DAI). The use of agents is consistent with the need for autonomy and communication in this domain.

The integration of the user as the supply chain engineer was realized by the configurational concept of the MAS. The user can define the MAS by using a graphical editor, without the knowledge of agent programming. He can shape the organizational structure of the network as well as the behavior of each agent. The knowledge representation for the behavior configuration follows the easy accessible rule syntax approach.

Future work has to consider the evaluation in a physical environment as well as a comparison to the results of a computer simulation. Furthermore, the integration of different AI techniques like Genetic Algorithms or Bayesian learning have to be evaluated for an improvement of the quality of the planning solution.

7. Related Work

There is other work in the Multi-Agent based Supply Chain Management. Existing appendages of Multi-Agent Systems in the field of production and logistics can be divided into internal and interplant concepts. While the internal concepts do not contain inter-organizational coordination, the interplant concepts aim exactly at that.

The Agent.Enterprise project [8] aims to establish a Multi-Multi-Agent System (MMAS). The Multi-Agent Systems are situated on the different intra- and inter-organizational levels of the supply chain. Therefore, the integration of the various MAS to a MMAS is essential. DISPOWEB is the inter-organizational MAS, focusing on a negotiation between the different companies, considering prices and time aspects. These results are given to the intra-organizational MAS like KRASH [14], IntaPS [18] or FABMAS [12]. These MAS are responsible for the production planning and control of different types of manufacturing environments.

8. References

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